KNECK OFFICIAL

Computer Studies — Paper 1

451/1 • ORIGINAL REVISION SET 01 • 2 hours 30 minutes • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all of Section A and ANY THREE questions in Section B. Use clear pseudocode and labelled diagrams. No computer is required for this theory paper.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

40 marks; all questions.

1. Distinguish hardware from software; give one example of each. [4 marks]

2. State two functions of an operating system and explain one. [4 marks]

3. Convert binary 101101 to decimal and decimal 26 to binary, showing working. [4 marks]

4. Explain the difference between RAM and secondary storage. [4 marks]

5. State two benefits and two risks of networking school computers. [4 marks]

6. Distinguish a primary key from a foreign key in a relational database. [4 marks]

7. Give two signs of a phishing message and two safe responses. [4 marks]

8. Describe four stages of the data-processing cycle. [4 marks]

9. Distinguish validation from verification and give an example of each. [4 marks]

10. Explain two accessibility features that help users with disabilities. [4 marks]

SECTION B

Choose any THREE; 20 marks each.

11. A school records Subject, Score and Grade. (a) Write pseudocode to input 8 scores, reject scores outside 0–100, calculate mean, and output Pass if mean≥50, otherwise Retry. (10) (b) Trace it for 40,50,60,70,80,90,30,60. (4) (c) Distinguish syntax, logic and runtime errors with examples. (6) [20 marks]

12. (a) Design a database for Students and ExamDownloads: fields/types/keys and relationship. (8) (b) Explain four methods of protecting school data. (8) (c) State two consequences of poor backup and two features of a tested backup plan. (4) [20 marks]

13. (a) Explain roles of router, switch, access point and firewall. (8) (b) Compare star and bus topology in three paired points. (6) (c) Describe three measures to secure school Wi-Fi. (6) [20 marks]

14. (a) Explain five phases of systems development. (10) (b) Compare direct and parallel changeover. (4) (c) State three testing types and give purpose of each. (6) [20 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (4 marks) Physical components versus programs/instructions (2); matched examples e.g. keyboard and operating system (2).

2. (4 marks) Memory/process/file/device/security/user-interface management, any two (2); one function explained accurately (2).

3. (4 marks) 101101_2=32+8+4+1=45 (2). 26=16+8+2=11010_2 (2).

4. (4 marks) RAM volatile working memory (2); secondary storage non-volatile long-term data retention (2).

5. (4 marks) Resource sharing,communication,central management (two); malware spread,unauthorised access,dependency/privacy risk (two).

6. (4 marks) Primary uniquely identifies each record and is non-null (2); foreign references key in another related table/enforces relationship (2).

7. (4 marks) Unexpected urgency/suspicious sender or links/credential request (two); independently verify/report/do not enter secrets/open unsafe attachments (two).

8. (4 marks) Input,processing,output,storage; one accurate role for each.

9. (4 marks) Validation checks acceptability e.g. range (2); verification checks accurate transcription e.g. double entry (2).

10. (4 marks) Screen reader with semantic text; keyboard navigation; captions; contrast/zoom; two feature-plus-benefit pairs.

SECTION B

11. (20 marks) a Initialise sum/count, loop 8 valid values, validation/re-entry, accumulation, mean, conditional output; allocate 10 to complete correct logic. b Sum480,mean60,Pass with trace evidence (4). c Syntax invalid grammar; logic wrong formula; runtime e.g. missing file/division by zero, three explained examples (6).

12. (20 marks) a Students StudentID PK/name/class; Downloads DownloadID PK,StudentID FK,ExamID,date; suitable types and 1:many relation (8). This is a hypothetical school database, not a requirement for public revision-site logins. b Least privilege,encryption,patching,authentication,logging,training four explained (8). c Data loss/disruption; offline/versioned copies and restore testing (4).

13. (20 marks) Router connects networks; switch forwards frames within LAN; AP wireless connection; firewall filters traffic, 2 each (8). Central device vs shared backbone, fault effects,cabling/performance three paired points (6). WPA2/3 strong credentials,guest isolation,updates/access controls, three

explained (6).

14. (20 marks) Problem analysis,requirements,design,implementation/testing,deployment/maintenance, five explained (10). Direct switch cheap/fast but risk; parallel old+new more cost but fallback (4). Unit,integration,system,acceptance,security/usability any three with purpose (6).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf